effective **`requirements.txt`** file for a **Scikit-learn** network intrusion detector needs the entire data science workflow.(like NSL-KDD, UNSW-NB15, or CICIDS2017), preprocessing network traffic features, handling class imbalances, and evaluation. [1, 2, 3]

Python `requirements.txt`

text

```
# Core Data Handling & Numerical Operations
numpy>=1.24.0
pandas>=2.0.0

# Machine Learning Core (Scikit-learn)
scikit-learn>=1.3.0

# Class Imbalance Handling (Essential for Network Attacks)
imbalanced-learn>=0.11.0

# Specialized Gradient Boosting (Optional but highly recommended for
tabular data)
xgboost>=1.7.0
lightgbm>=4.0.0

# Public Dataset Retrieval & Evaluation Utilities
scipy>=1.10.0
joblib>=1.3.0

# Visualization & Analysis (For confusion matrices, ROC curves)
matplotlib>=3.7.0
seaborn>=0.12.0
```

Use code with caution.

Component Breakdown & Dependency Context

Dependency	**Purpose in Network**

	Intrusion Detection
`pandas` & **`numpy`**	Parsing high-volume network packet captures, processing logs, and handling feature matrices.
`scikit-learn`	Core library providing the pipeline architecture (`MinMaxScaler`, `StandardScaler`), feature selection (`SelectKBest`), and ML classifiers like

	Random Forest, **SVM**, and **Logistic Regression**.
`imbalanced-learn`	Critical for over-sampling rare attacks (like U2R or R2L) relative to normal traffic using techniques like **SMOTE** or **ADASYN**.
`xgboost` / `lightgbm`	Standard additions because ensemble gradient boosting frequently yields the highest accuracy on tabular

	network security datasets.
`seaborn` / `matplotlib`	Necessary for graphing training curves and rendering confusion matrices to assess false-alarm vs. detection rates